Shift-Share Designs and Event Studies

Topics in Macroeconomics – Week 2

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PhD elective, 2026/27

Bank of Portugal & Católica Lisbon

Today

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Material draws on Borusyak, Hull & Jaravel (2025), Hull (Applied Econometrics, ch. 8) and Borusyak (ARE 213 lectures).

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$\Rightarrow$ The answer to the second depends on the answer to the first

Why shift-share?

The shift-share instrument

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- Where it comes from: the treatment itself has this form, $x_\ell = \sum_n s_{\ell n}\, x_{\ell n} \rightarrow$ replace the local growth rates $x_{\ell n}$ by common ones g_n

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Study	Unit ℓ	Shares $s_{\ell n}$	Shifts g_n
Bartik (1991)	region	industry employment	national industry growth
Card (2009)	city	migrants by origin	new migrants by origin
Autor, Dorn & Hanson (2013)	comm. zone	industry employment	Chinese imports elsewhere
Nakamura & Steinsson (2014)	state	procurement sensitivity	national military build-up
Greenstone, Mas & Nguyen (2020)	county	bank market shares	bank lending shocks
Acemoglu & Restrepo (2020)	comm. zone	industry employment	robots, Europe

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$\Rightarrow$ Most cross-sectional macro evidence is a shift-share design, named or not

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- **Findings:** more exposed areas saw higher unemployment, lower participation, lower wages and rising transfer payments; import competition explains about a quarter of the decline in US manufacturing employment
- **Why it matters:** launched the "China shock" literature (labour markets, politics, innovation) and became the textbook shift-share design: GPSS, AKM and BHJ all re-analyse it

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- **Ideal experiment:** assign Chinese productivity gains at random across industries; regions differ in how exposed they are through their industry mix

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- **Treatment and instrument** (shares lagged a decade so they predate the shock):

$$x_{\ell t} = \sum_n \frac{L_{\ell nt}}{L_{\ell t}} \cdot \frac{\Delta M_{nt}^{US}}{L_{nt}}, \qquad z_{\ell t} = \sum_n \underbrace{\frac{L_{\ell n,t-10}}{L_{\ell,t-10}}}_{s_{\ell nt}} \cdot \underbrace{\frac{\Delta M_{nt}^{OTH}}{L_{n,t-10}}}_{g_{nt}}$$

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- **Design:** stacked first differences, 1990–2000 and 2000–07 (per decade); 2SLS weighted by CZ population, SEs clustered by state, start-of-period CZ controls

The instrument

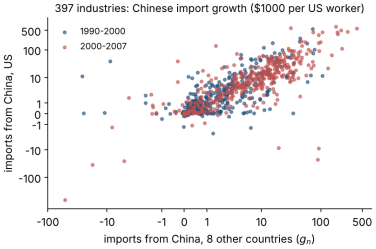
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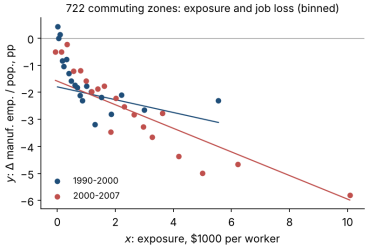
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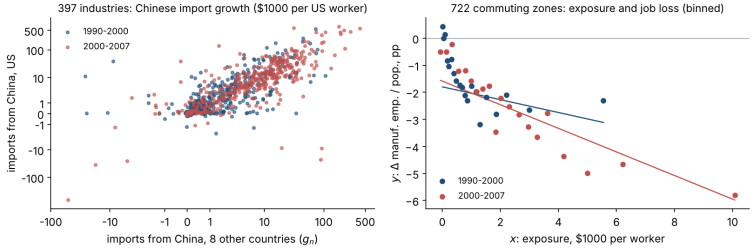
$\Rightarrow$ A translation device: industry-level shocks, region-level outcomes

The data: shifts and outcomes



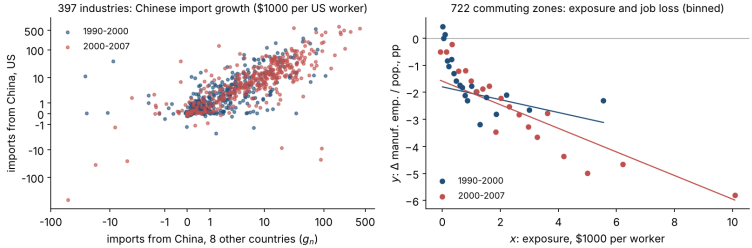


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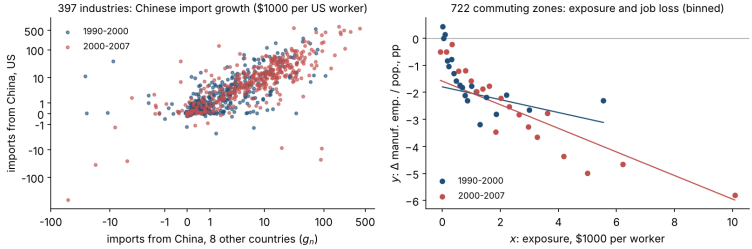
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$\Rightarrow$ Right panel = the simple regression (S1); the instrument keeps only the exposure driven by the left panel

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2000–07	California CZ	Michigan CZ
Manufacturing share, $S_{\ell t} = \sum_n s_{\ell nt}$	0.31	0.34
Largest industries	chips, computers	office furniture, car parts
Their shocks g_{nt}	42, 189	0.6, 8.5
Instrument $z_{\ell t} = \sum_n s_{\ell nt}\, g_{nt}$	10.7	2.9
of which computers alone	4.9	0.0
Exposure $x_{\ell t}$	7.3	3.1
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$\Rightarrow$ The instrument asks: did CZs specialised in the industries where China took off lose more jobs?

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- **Shift-share IV** (theory, then the example):

  S1 OLS S2 The ADH instrument S3 Incomplete shares S4 Standard errors
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$\Rightarrow$ Every number and figure comes from `week2_simple.py` (one cell per step) and `week2_fast.py`

Two routes to identification

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$\Rightarrow$ Pick one and say which: controls, inference and diagnostics differ

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$$\hat{\beta} = \sum_k \hat{\alpha}_k \hat{\beta}_k, \qquad \hat{\beta}_k = \frac{s_k' M y}{s_k' M x}, \qquad \hat{\alpha}_k = \frac{g_k \, s_k' M x}{\sum_j g_j \, s_j' M x}$$

  - M: residualise on the controls; $\hat{\alpha}_k$ = Rotemberg weights (sum to one, can be negative)
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- **Diagnostics:** which shares drive $\hat{\beta}$ (S6), balance on those shares, over-identification

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- Write the covariance at the shock level (weights w_ℓ, shock weight $s_n = \sum_\ell w_\ell s_{\ell n}$):

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$\Rightarrow$ The natural route for the China shock: many industries, shocks driven by Chinese supply

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- A constant at the shock level becomes the control $S_{\ell} = \sum_{n} s_{\ell n}$ in the CZ regression (next slide)

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$\Rightarrow$ Exposure to random shocks is not random when exposure itself is non-random

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$\Rightarrow$ **The route decides the SEs:** SEs redraw the random part

- Route 2, random shocks: CZs with similar industries move together $\Rightarrow$ exposure-robust
- Route 1, random shares, fixed shocks: CZs are the independent units $\Rightarrow$ conventional

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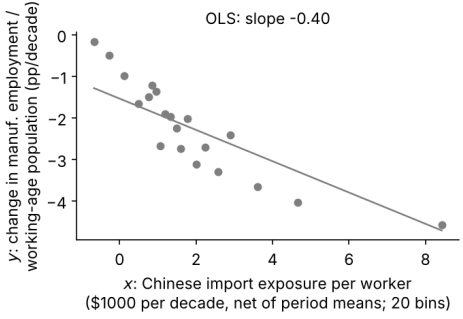
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$\Rightarrow$ For shares: tailored shares, Rotemberg weights, balance on top shares, over-identification

The China shock, step by step

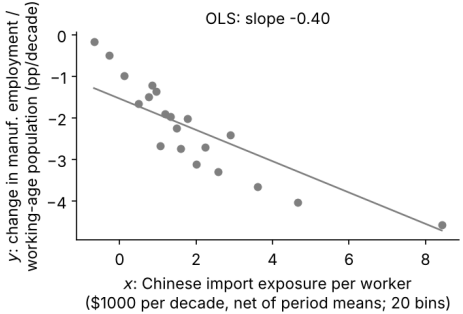
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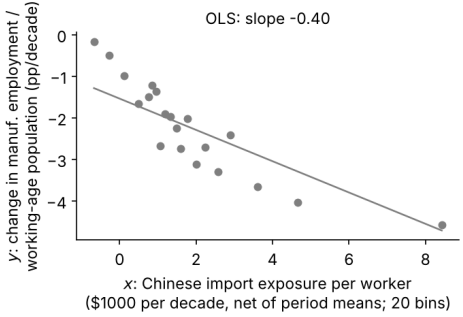
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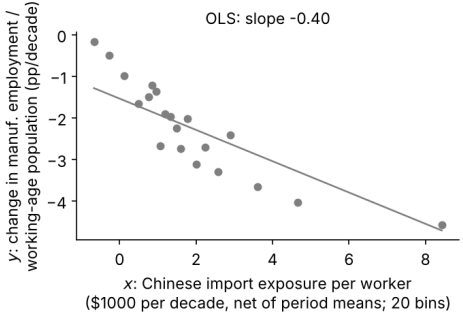
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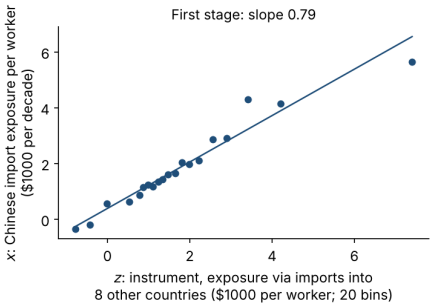


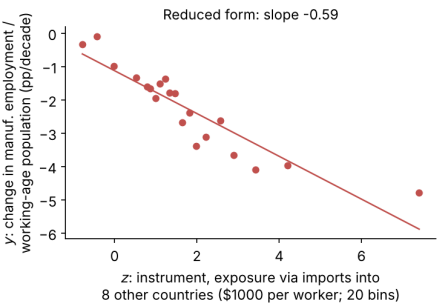
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$\Rightarrow$ Biased towards zero if US demand shocks raise imports *and* jobs

S2: the ADH instrument

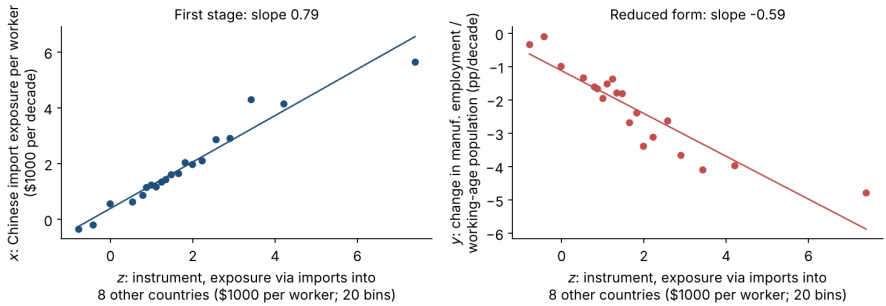
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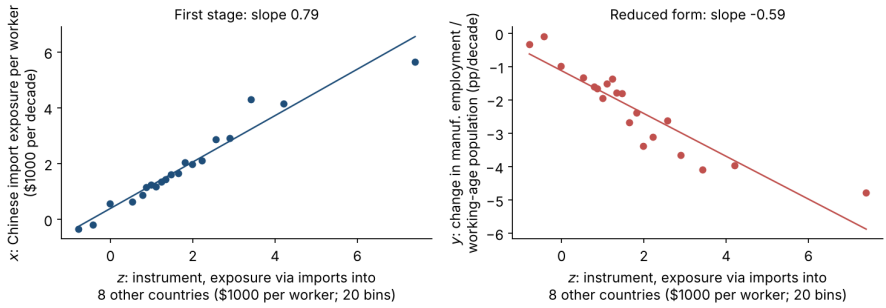
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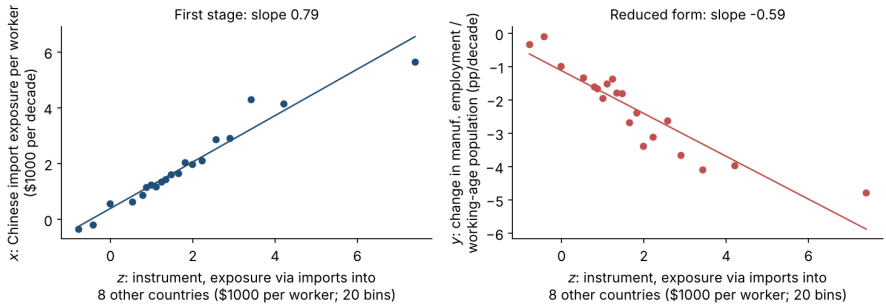
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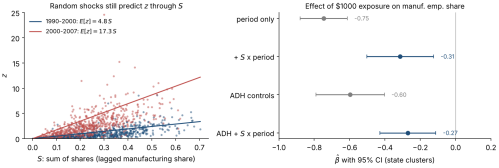


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$\Rightarrow$ IV $>$ OLS in absolute value, as the demand story predicts

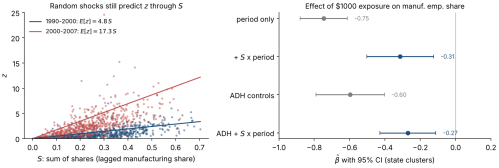
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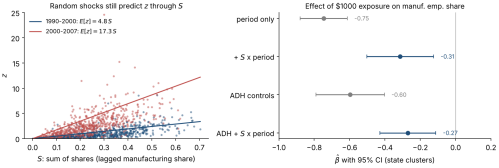
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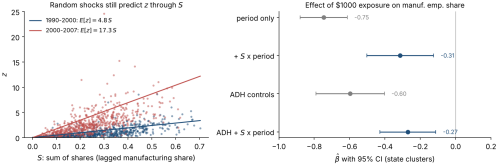
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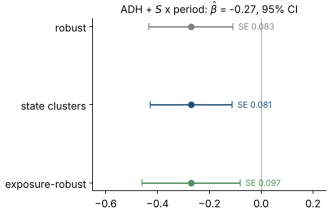
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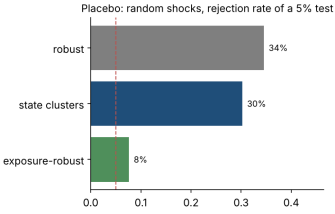
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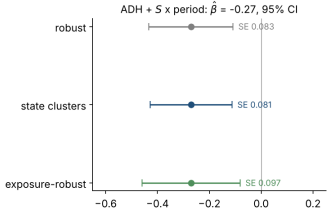
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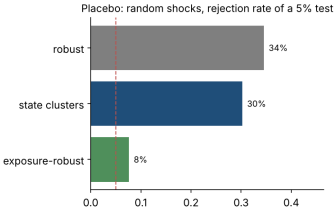
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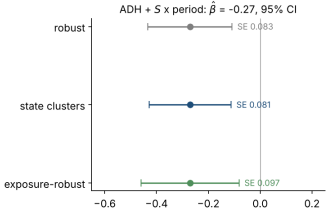
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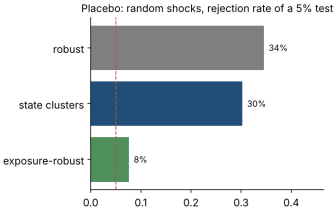




- How precise is the estimate? The randomness is in 794 industry shocks, not in 1444 CZs

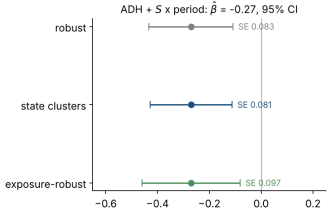
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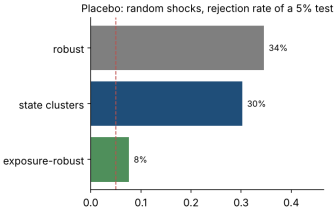




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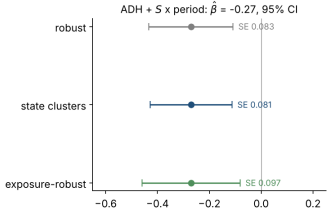
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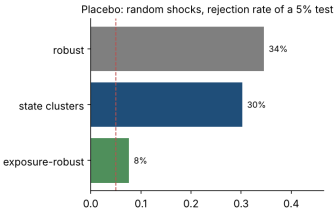




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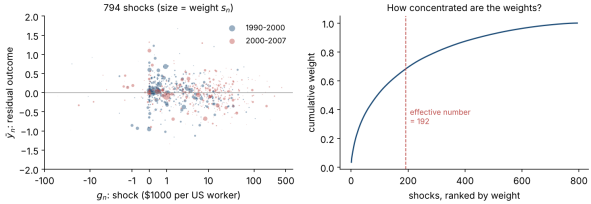




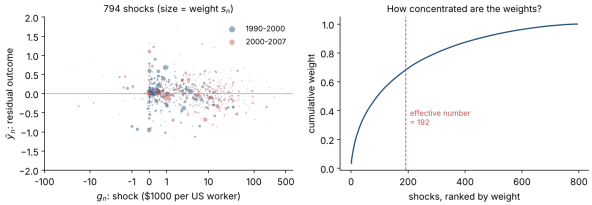
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$\Rightarrow$ Our estimate: SE 0.081 $\rightarrow$ 0.097, still significant. In other designs the correction can kill the result

S5: the shock-level view

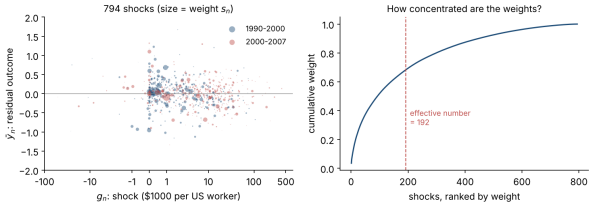


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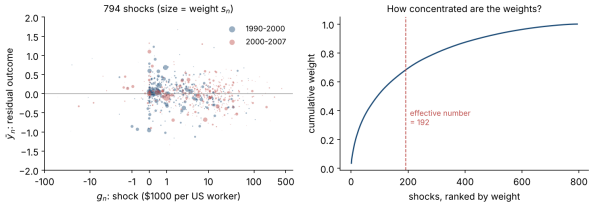
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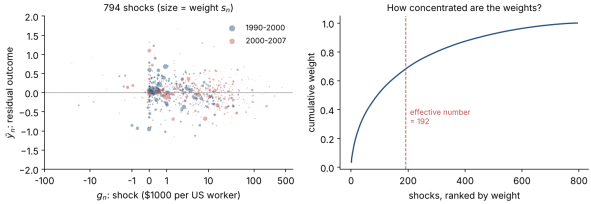
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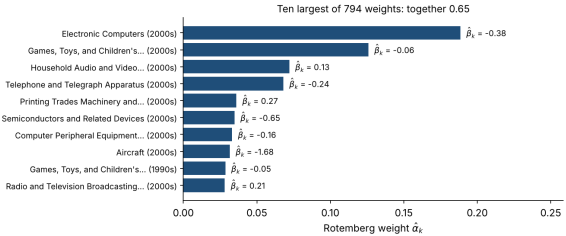
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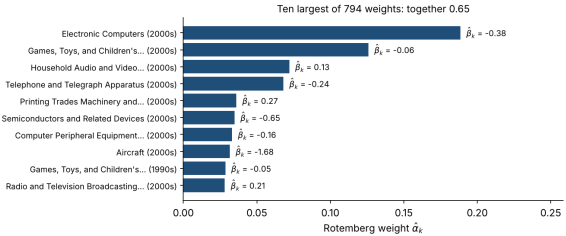
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$\Rightarrow$ Many small shocks, none dominant: A2 is plausible. With a dozen, a few industries would decide $\hat{\beta}$

S6: Rotemberg weights

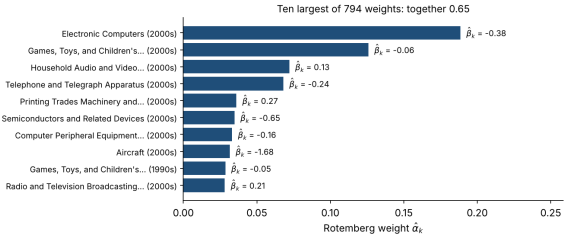


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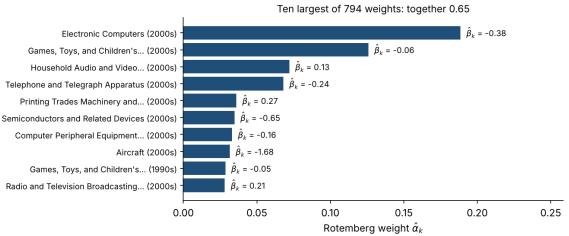
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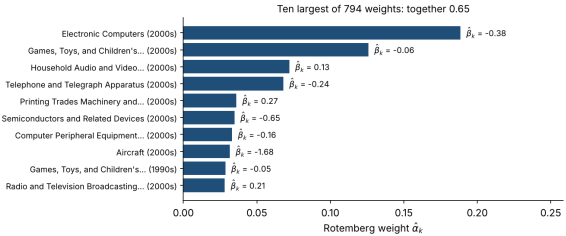
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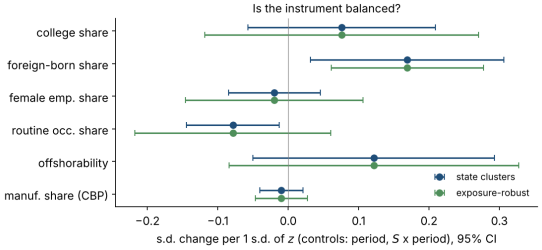


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$\Rightarrow$ Always ask which part of the data drives the number. Check: drop computers $\Rightarrow -0.26$, robust

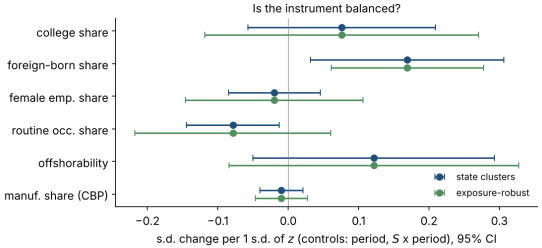
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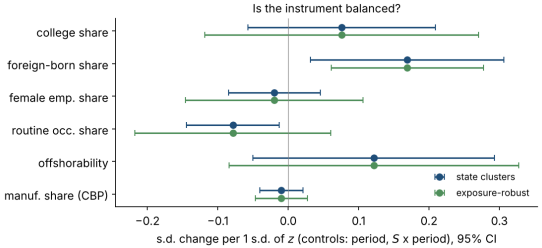
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$\Rightarrow$ A reason to keep ADH's controls; check also shock-level balance (industry skill, wages, capital intensity) and pre-trends

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- Ways forward:
  - A model disciplined by the cross-sectional estimate
  - Aggregate time-series evidence for the intercept (Week 1)

Event studies with staggered adoption

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- **Link to shift-share:** same panel logic, units with different exposure, net of common time effects
  - Shift-share: how much a unit is exposed. Staggered: when
  - Same question as S6: *what does the regression actually compare?*
  - Shift-share: set by random shocks
  - Staggered TWFE: set by timing, already-treated units act as controls

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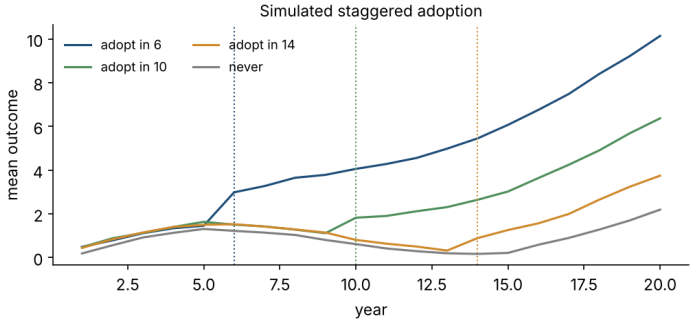
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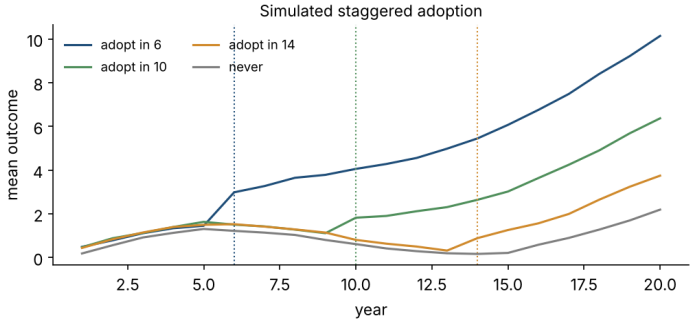
- The effect τ_{it} can differ by cohort and grow with time since adoption $k = t - E_i$
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$$\begin{aligned} \text{static TWFE:} \quad & Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + e_{it} \\ \text{event study:} \quad & Y_{it} = \alpha_i + \lambda_t + \textstyle\sum_{k \neq -1} \beta_k \, 1\{t - E_i = k\} + e_{it} \end{aligned}$$

E0: a simulated panel where we know the truth

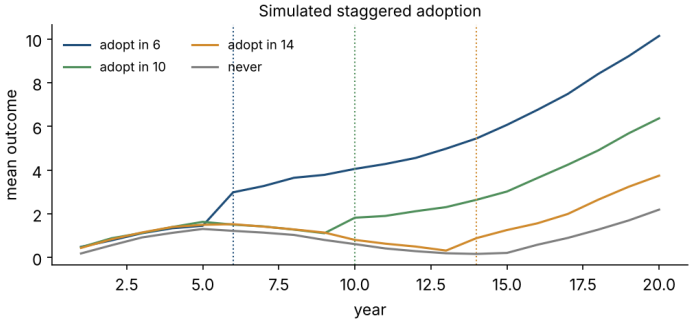


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- Effect grows with time since adoption and is larger for early adopters: $\tau_{it} = c_{E_i}\,(1 + 0.3\,k)$, $c = 1.5,\ 1,\ 0.5$

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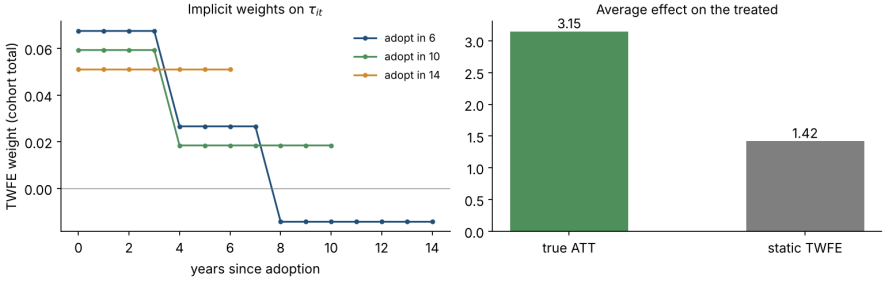
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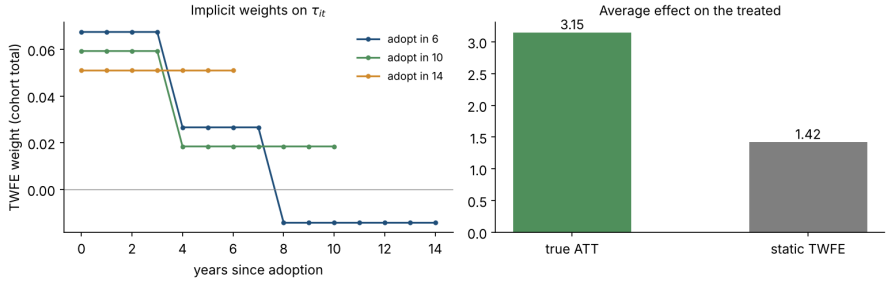
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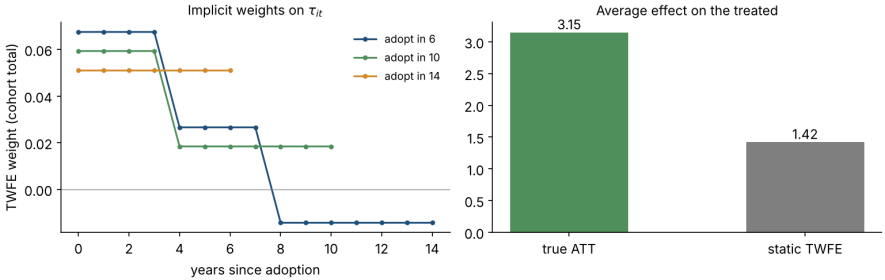


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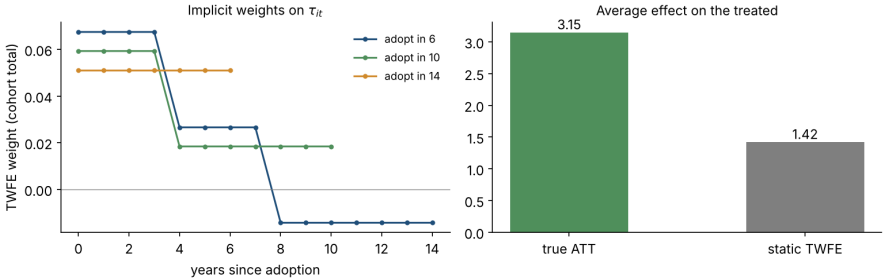
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$\Rightarrow$ TWFE overweights short-run effects and puts negative weight on the largest ones

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- **Fix** : compare treated cells only with untreated ones – the imputation estimator
  - Estimate on clean comparisons; test pre-trends separately on untreated cells

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- **Pre-trends:** regress untreated Y_{it} on unit and year effects plus lead dummies – a separate test

LP-DiD: Week 1's local projection as a DiD

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$$Y_{i,t+h} - Y_{i,t-1} = \beta_h \, \Delta D_{it} + \delta_t^h + e_{it}^h, \qquad h = 0, 1, \ldots, H$$

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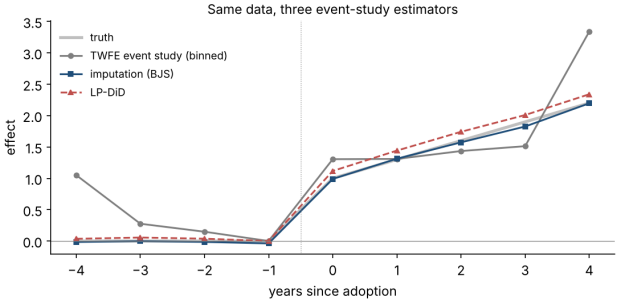
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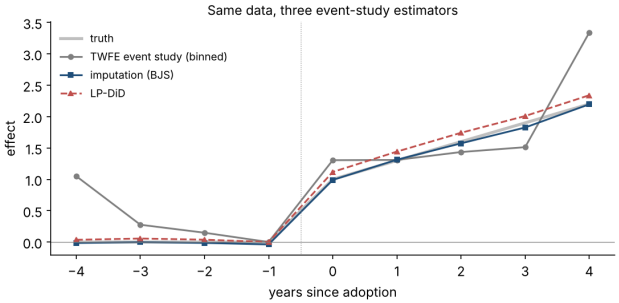
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$\Rightarrow$ Clean-control restriction = the identifying assumption, like the controls in an LP

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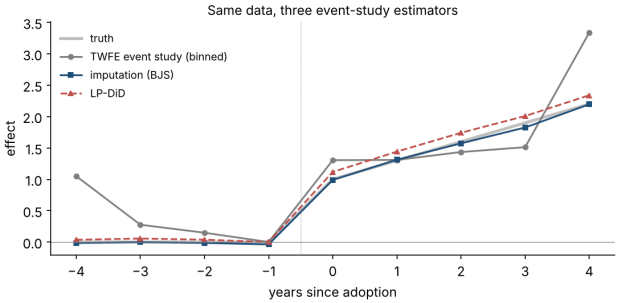


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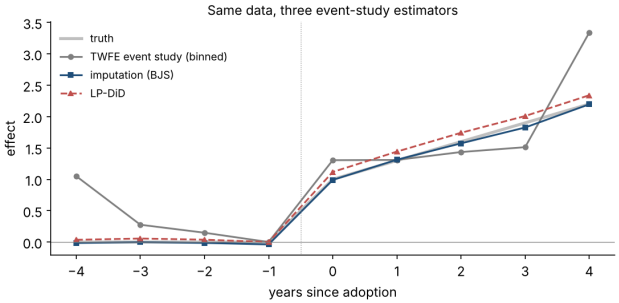
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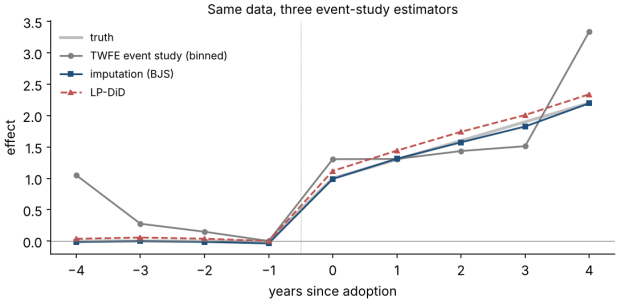
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$\Rightarrow$ Same data, same parallel trends: only the comparisons differ

How it fits together

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$\Rightarrow$ Ask every design: what is random, how many draws of it are there, what does the regression weight?

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- Next: implementation: the same steps in code

Implementation

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- **Exercise** ($\sim$25 minutes): `exercise_ADH_student.py`, six short parts with `check()` lines

The instrument in code: a merge and a group-by

```
shares = pd.read_stata("data/Lshares.dta")            # czone, year, sic87dd, ind_share
shocks = pd.read_stata("data/shocks.dta")             # year, sic87dd, g

m = shares.merge(shocks[["year", "sic87dd", "g"]], on=["year", "sic87dd"])
m["sg"] = m.ind_share * m.g                           # s_ln * g_n
inst = m.groupby(["czone", "year"]).agg(z=("sg", "sum"), S=("ind_share", "sum")).reset_index()
df = df.merge(inst, on=["czone", "year"])
df["S_90"], df["S_00"] = df.S * (1 - df.t2), df.S * df.t2     # S x period

ADH = "l_shind_manuf_cbp + l_sh_popedu_c + ... + C(division)"
res = IV2SLS.from_formula(f"d_sh_empl_mfg ~ 1 + t2 + S_90 + S_00 + {ADH} + [shock ~ z]",
                          df, weights=df.weights).fit(cov_type="clustered", clusters=df.statefip)
print(res.params["shock"])                            # -0.270
```

Exposure-robust SEs in code: the shock-level IV

```
def shock_level(cols, controls, data):
    r = data[["czone", "year", "weights"]].copy()
    for c in cols:                                    # 1. residualise on the CZ controls
        r[c] = smf.wls(f"{c} ~ {controls}", data, weights=data.weights).fit().resid
    sl = shares.merge(r, on=["czone", "year"])        # 2. attach residuals to shares
    sl["ws"] = sl.weights * sl.ind_share
    for c in cols:
        sl[c] = sl[c] * sl.ws
    out = sl.groupby(["year", "sic87dd"])[cols + ["ws"]].sum()     # 3. average by shock
    out[cols] = out[cols].div(out.ws, axis=0)
    out = out.reset_index().merge(shocks[["year", "sic87dd", "g"]], on=["year", "sic87dd"])
    out["sic3"] = (out.sic87dd // 10).astype(int)
    return out

sh = shock_level(["d_sh_empl_mfg", "shock"], PREF, df)              # 4. IV across shocks
r = IV2SLS.from_formula("d_sh_empl_mfg ~ 0 + C(year) + [shock ~ g]", sh, weights=sh.ws) \
          .fit(cov_type="clustered", clusters=sh.sic3)                # -0.270 (0.097)
```

Event studies in code: imputation and LP-DiD

```
# Imputation: unit and year effects from untreated cells only, then average by k
imp = smf.ols("Y ~ C(i) + C(t)", panel[panel.D == 0]).fit()
panel["tau_hat"] = panel.Y - imp.predict(panel)
es_imp = [panel[panel.k == k].tau_hat.mean() for k in range(0, 9)]

# LP-DiD: one long-difference regression per horizon, newly treated vs clean controls
def lpdid(h):
    rows = []
    for t in range(2, T + 1):
        if not 1 <= t + h <= T:
            continue
        newly = (wideD[t] == 1) & (wideD[t - 1] == 0)
        clean = wideD[max(t, t + h)] == 0
        keep = newly | clean
        rows.append(pd.DataFrame({"dy": wideY[t + h][keep] - wideY[t - 1][keep],
                                  "dD": newly[keep].astype(int), "t": t}))
    return smf.ols("dy ~ dD + C(t)", pd.concat(rows)).fit().params["dD"]
```

$\Rightarrow$ Same loop over h as the Week 1 LP; only the sample changes with h

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`exercise_ADH_student.py`: fill in the `# TODO` lines; each `check()` prints ✓ when your number matches.

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- **Take-home** (due Week 3): the BHJ checklist for the China shock; spillovers with exposure-robust SEs; event-study estimators with cohort-specific constant effects

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- **Next week:** financial crises and policy in open economies – sudden stops, collateral constraints, macroprudential policy